MIL-P-44147(GL)

28 March 1985

MILITARY SPECIFICATION

PUDDING BAR - (OPERATIONAL RATION COMPONENT)

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers pudding bars for use by the Department of Defense as a component of the Food Packet, Assault.

1.2 Classification. The pudding bars shall be the following types as specified (see 6.1):

Type I - vanilla flavored

Type II - chocolate flavored

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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SPECIFICATION

MILITARY

MIL-P-44146 - Packaging of Food Packet Assault Components in Flexible Pouches

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Instant Nonfat Dry Milk

(Application for copies should be addressed to the Director, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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National Academy of Sciences

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209)

THE UNITED STATES PHARMACOPOEIAL CONVENTION (USP)

Pharmacopoeia of the United States

(Application for copies should be addressed to the United States Pharmacopoeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

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3.2.1 Sugar, super fine, granulated. Super fine, granulated sugar shall be white, refined, and shall contain not less than 99.5 percent sucrose, not more than 0.04 percent ash, and not more than 0.10 percent moisture. Not more than 1 percent of the sugar shall be retained on a U.S. Standard No. 40 sieve, not more than 20 percent shall pass through a U.S. Standard No. 140 sieve, and not more than 5 percent shall pass through a U.S. Standard No. 200 sieve. The sugar shall not contain lumps larger than ½-inch in their greatest diameter that cannot be broken by light finger pressure.

3.2.2 Milk, nonfat, dry, instant. Instant nonfat dry milk shall meet the requirements for U.S. Extra Grade, Low Heat, as defined in the U.S. Standards for Instant Nonfat Dry Milk.

3.2.3 Starch, instant. Instant starch shall be a waxy maize starch that will produce immediate high viscosity with extreme smoothness. The starch shall have a composition equivalent to Pure-Flo-F manufactured by National Starch and Chemical Corporation.

3.2.4 Cocoa powder. Cocoa powder shall be made from the nibs of domestically roasted, mature, well-fermented cocoa beans which have been properly dried. Either the nibs or the liquor from the nibs shall be mildly alkalized. The pH shall be not less than 6.0 nor more than 7.9. The cocoa powder shall contain at least 12.0 percent cocoa butter. Chemically extracted cocoa, in whole or in part, shall not be acceptable. Not less than 99.0 percent of the powder shall pass through a U.S. Standard No. 100 sieve. The cocoa shall be salmonella negative in 125 grams.

3.2.5 Dextrose. Dextrose shall be free-flowing dextrose hydrate powder.

3.2.6 Shortening, powdered. Powdered shortening shall be spray dried, free-flowing, and free of burnt or scorched aromas. The shortening shall be composed of partially hydrogenated vegetable oil, sodium caseinate, corn syrup solids, soy protein isolate, dipotassium phosphate, sodium silicoaluminate and tricalcium phosphate; with or without antioxidant and shall have the following typical analysis:

Total oil percent	75 ± 2.5
Protein, percent	5.5 ± 0.5
pH	7.5 ± 0.5
Moisture percent	1.5 ± 0.5
Carbohydrate, percent	16.0 ± 1.0
Ash, percent	2.0 ± 1.0

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3.2.7 Tetrasodium pyrophosphate. Tetrasodium pyrophosphate shall comply with the requirements of the Food Chemicals Codex, shall be water soluble, and shall be pudding grade.

3.2.8 Calcium acetate. Calcium acetate shall comply with the requirements of the Food Chemicals Codex.

3.2.9 Disodium phosphate. Disodium phosphate shall comply with the requirements of the Food Chemicals Codex.

3.2.10 Salt. Salt shall be non-iodized, white, refined sodium chloride, with or without anti-caking agents.

3.2.11 Sodium caseinate. Sodium caseinate shall be spray dried, food grade sodium caseinate made from fresh skim milk.

3.2.12 Glycerine. Glycerine shall comply with the requirements of the Food Chemicals Codex.

3.2.13 Titanium dioxide. Titanium dioxide shall comply with the requirements of the Food Chemicals Codex, and shall be water dispersible.

3.2.14 Vanilla cream flavor. Vanilla cream flavor, shall be PFW No. 610346U manufactured by PFW Division of Hercules, Inc., or equivalent.

3.2.15 Vanilla flavor, artificial. Artificial vanilla flavor shall be PFW No. 650 204 U manufactured by PFW Division of Hercules, Inc., or equivalent.

3.2.16 Yellow no. 5 color. Yellow no. 5 color shall comply with the requirements of the Food, Drug and Cosmetic Act and regulations promulgated thereunder.

3.2.17 Yellow no. 6 color. Yellow no. 6 color shall comply with the requirements of the Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.3 Preparation and processing. Ingredient storage and preparation and processing of the pudding bars shall be at a relative humidity not to exceed 40 percent.

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3.3.1 Preparation of the vanilla flavored pudding mix.

3.3.1.1 Formulation of the vanilla flavored pudding mix. The vanilla flavored pudding mix shall be formulated as follows:

Ingredient	Parts by weight
Sugar, super fine, granulated	52.17 to 55.67
Milk, nonfat, dry, instant	23.00
Starch, instant	12.00 to 13.00
Shortening, powdered	3.50 to 5.50
Dextrose	3.00
Tetrasodium pyrophosphate	0.85
Disodium phosphate	0.55
Salt	0.40
Sodium caseinate	0.35
Glycerine	0.25 to 0.75
Titanium dioxide	0.15
Vanilla cream flavor	0.03
Vanilla flavor, artificial	0.02
Yellow no. 5 color	0.007
Yellow no. 6 color	0.002

3.3.1.2 Mixing of the vanilla flavored pudding. The vanilla flavored pudding shall be mixed as follows:

a. The mixer shall be started (medium speed) and the sugar and color shall be added. The sugar and color shall be blended for 2 minutes.

b. The glycerine shall be slowly added and blending shall continue for an additional 3 to 4 minutes.

c. The powdered shortening shall be added and blending shall continue for an additional 1 to 2 minutes.

d. The remaining dry ingredients except the nonfat dry milk shall be added and blending shall continue for an additional 3 to 4 minutes.

e. The nonfat dry milk shall be added and blending shall continue for an additional 3 to 4 minutes.

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3.3.2 Preparation of the chocolate flavored pudding mix.

3.3.2.1 Formulation of the chocolate flavored pudding mix. The chocolate flavored pudding mix shall be formulated as follows:

Ingredient	Parts by weight
Sugar, super fine, granulated	45.51 to 42.01
Milk, nonfat, dry, instant	25.00
Starch, instant	10.00 to 11.00
Cocoa powder	8.50 to 9.50
Dextrose	4.00
Shortening, powdered	3.50 to 4.50
Tetrasodium pyrophosphate	1.60
Calcium acetate	0.80
Salt	0.40
Sodium caseinate	0.35
Glycerine	0.25 to 0.75

3.3.2.2 Mixing of the chocolate flavored pudding. The chocolate flavored pudding shall be mixed as follows:

a. The mixer shall be started (medium speed) and the sugar added.

b. The glycerine shall be slowly added to the sugar. The glycerine and sugar shall be blended for 3 to 4 minutes.

c. The powdered shortening shall be added and blending shall continue for an additional 1 to 2 minutes.

d. The cocoa powder shall be added and blending shall continue for an additional 3 to 4 minutes.

e. The remaining dry ingredients, except the nonfat dry milk, shall be added and blending shall continue for an additional 3 to 4 minutes at slow speed.

f. The nonfat dry milk shall be added and blending shall continue for an additional 3 to 4 minutes.

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3.3.3 Compression of pudding mix. The vanilla flavored pudding mix or chocolate flavored pudding mix shall be compressed into bars 1-inch wide by 3-inches long by not more than ½-inch thick so that the bars will meet the requirements specified in 3.4. The bars shall have edges and corners rounded to a nominal 1/8-inch curvature.

3.3.4 Pouch filling and sealing. Each pouch shall be filled and sealed such as to conform to the following requirements:

a. One bar shall be inserted into a mixing bag conforming to the requirements of MIL-P-44146.

b. The filled mixing bag shall be folded and fitted into a pouch.

c. The pouch shall be sealed in accordance with the requirements of MIL-P-44146.

3.4 Finished product. The finished product shall comply with the following requirements:

Dry mix product:

a. There shall be no foreign material, such as, but not limited to dirt, glass, paint, hard plastic, wood, insects, insect parts or noxious chemicals. There shall be no innocuous soft extraneous material (paper, string, etc.) or smears or stains (oil, grease, etc).

b. There shall be no foreign odor or flavor, such as, but not limited to stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. Each pouch shall contain one bar.

e. Each bar shall be 3 ± 0.2 by 1 ± 0.2 by not greater than ½ inches.

f. Each bar shall weigh not less than 30 grams and not greater than 40 grams.

g. Each bar shall have rounded edges and corners.

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h. The product shall be sufficiently compressed so that it can be packaged without breaking or crumbling.

i. The color of the vanilla flavored pudding bar shall be very light yellow. The color of the chocolate flavored pudding bar shall be light chocolate brown.

j. The flavor of the vanilla flavored pudding bar or chocolate flavored pudding bar shall be similar to commercial instant vanilla or chocolate pudding, respectively.

k. Moisture content of the dry mix product shall not exceed 2.0 percent.

l. The aerobic plate count shall be not greater than 75,000 per gram in four of five samples and not greater than 150,000 per gram in any sample.

m. The E. coli count shall be less than 3 per gram (no positives in the standard 3 tube MPN technique) in four of five samples and not greater than 20 per gram in any sample.

Prepared product

n. When the product is prepared as specified in footnote 3 of table I, the product shall exhibit a smooth texture, free from excessive lumps, similar to that of noncompressed instant vanilla or chocolate pudding.

3.4.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed, and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a

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product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this document. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition and other requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.5.2 In-process examination. In-process examination shall be performed to determine conformance to preparation, processing, handling, and filling requirements. Any nonconformance revealed by actual examination or by review of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch examination. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44146.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch and mixing bag. Any individual net weight of less than 30 grams or greater than 40 grams shall be scored as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 1 gram.

4.5.5 Product examination. The finished product shall be examined for the defects listed in table I. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

Table I. Product defects 1/ 2/

Category		Defect
Major	Minor	
151		Mixing bag missing.
152		Pudding bar missing.
		Applicable to dry mix product
	201	Broken or crumbled (more than 3 large pieces) bar.

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Table I. Product defects 1/ 2/ (con't)

Category		Defect
Major	Minor	
153		Color of vanilla flavored bar not very light yellow or color of chocolate flavored bar not light chocolate brown.
154		Flavor of vanilla flavored bar not similar to commercial instant vanilla pudding or flavor of chocolate flavored bar not similar to commercial instant chocolate pudding.
	202	Dimensions of bar not as specified.
	203	Edges and corners of bar not rounded.
		Applicable to prepared product 3/
155		Product does not exhibit a smooth texture, free from excessive lumps, similar to that of non-compressed instant vanilla or chocolate pudding.

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g. moldy, rancid, sour, stale, noxious chemicals etc.), foreign color or innocuous soft extraneous material, smears or stains (oil, grease, etc.) shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ Break the bar into five or six pieces of comparable size. Add 70 mL of 68° to 72°F water to the broken product and knead for 5 minutes.

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4.5.6 Moisture content testing. The finished product shall be tested for moisture content in accordance with the Moisture-Drying in Vacuum Method in the Meat section of the Meat and Meat Products chapter in the Official Methods of Analysis of the Association of Official Analytical Chemists, except that a temperature of 70°C for 16 hours under a pressure of 100 mm of mercury shall be used. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4k shall be considered a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.7 Microbiological testing. Five filled and sealed pouches of finished product shall be selected at random from the lot regardless of the lot size. The contents of each sample pouch shall be tested for aerobic plate count and E. Coli in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Microbiological Methods: Section: Examination of Frozen, Chilled, Precooked, or Prepared foods - Official First Action. The diluent shall be added to each sample of dry product and allowed to stand for 15 minutes before the blending of that sample. Continue as directed by AOAC. Any result not conforming to the requirements of 3.4 l and m shall be considered a test failure. Any test failure shall be cause for rejection of the lot.

4.5.8 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44146.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the applicable requirements of MIL-P-44146.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type of pudding bar required (see 1.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.4.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article

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should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Custodian:

Army - GL

Review activities:

Army - MD, TS
DP - SS

Preparing activity:

Army - GL

Project No. 8940 - A471

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-44147(GL)	Pudding Bar - (Operational Ration Component)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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